

AI • BACKEND API

AI BLOGNEST API

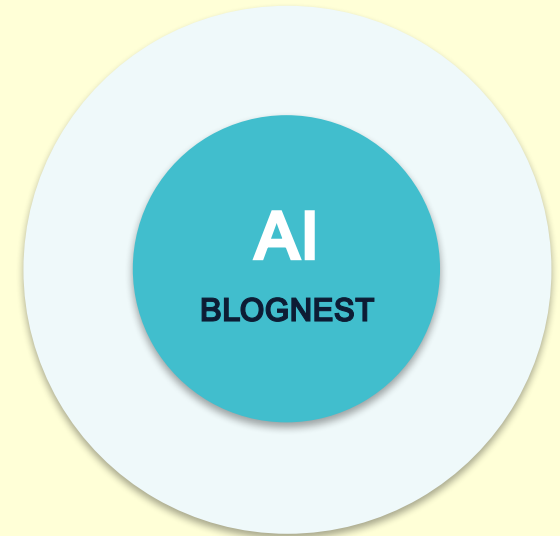
An AI-powered REST API for intelligent blog creation, content management, security and AI-assisted writing.

Node.js

Express.js

MongoDB

Gemini AI



PROJECT DEVELOPMENT TEAM

TEAM LEADER

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Project Overview

Why BlogNest / ContentGen is needed

Core Idea

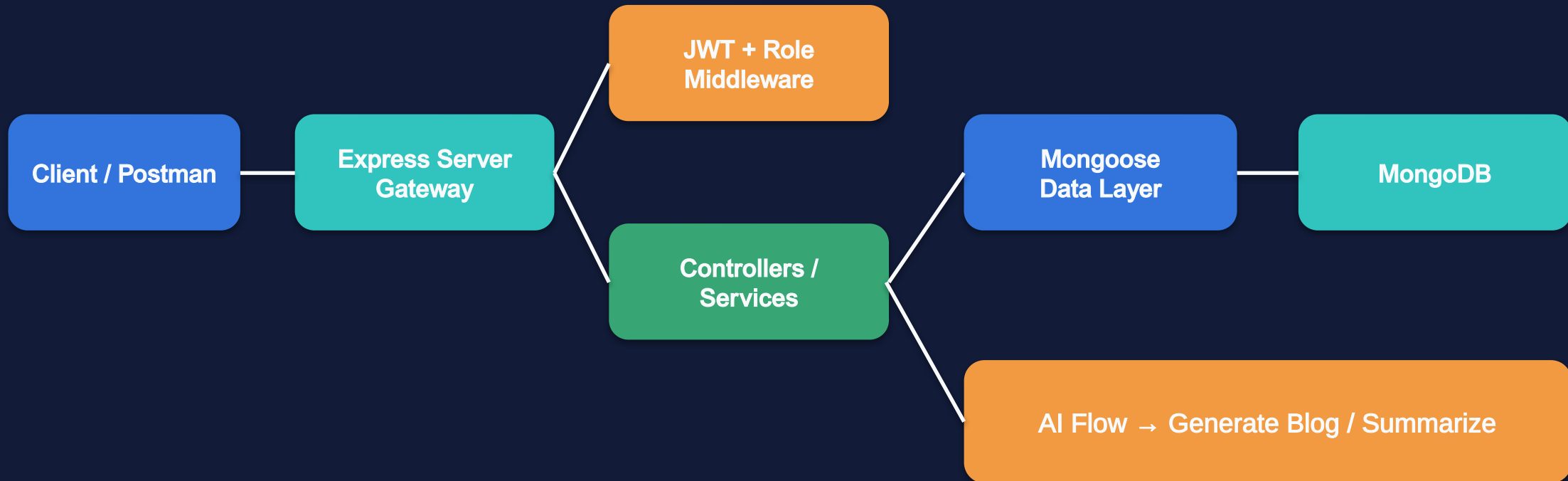
- Centralized platform for creating and managing blog content.
- Supports drafts, categories, tags, comments and media metadata.
- AI assistance for blog generation and content summarization.
- Role-based access for Admin, Editor, Author and Reader.
- Security includes JWT authentication, bcrypt hashing, validation and rate limiting.

Key Benefits

- Reduces manual effort in content creation.
- Improves approval and publishing workflows.
- Maintains structured and validated MongoDB data.
- Provides protected API endpoints for different user roles.
- Designed with a modular, scalable backend architecture.

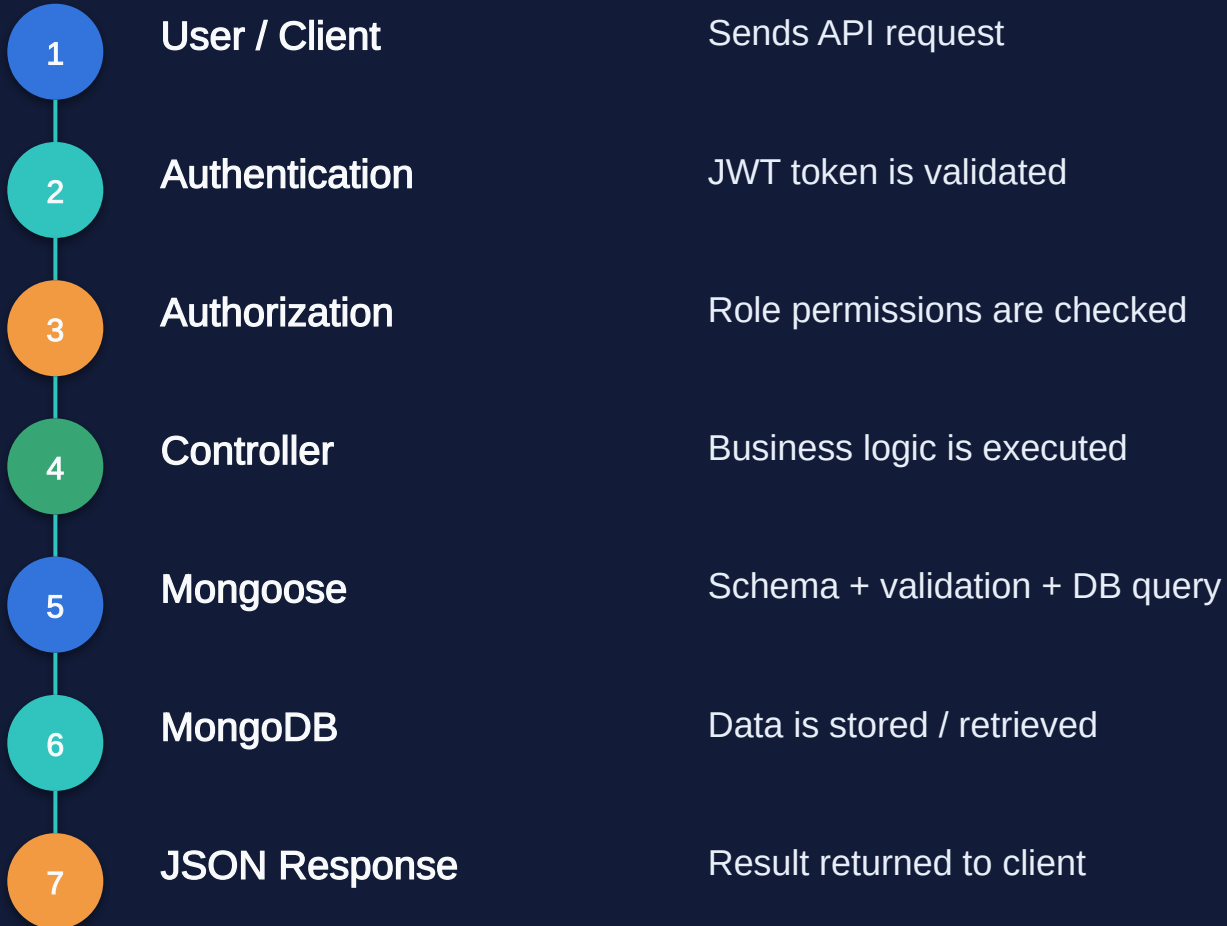
System Architecture

Layered backend architecture used by the application



User & API Flow

From request to database response



MVC + Database Design

How the backend separates responsibilities

MODEL

- Mongoose schemas define the data blueprint.
- Validation, defaults and indexing are applied before persistence.
- Collections include users, posts and comments.

User

Post

CONTROLLER

- Receives requests from routes.
- Checks constraints and delegates workflows.
- Returns standardized JSON responses.

Comment

VIEW / ROUTING

- Headless backend: no traditional UI templates.
- API routing layer connects paths to controllers.
- Examples: GET, POST, PUT and DELETE endpoints.

1 → Many

Roles

AI Features & Security

Intelligent automation with protected API access

AI CAPABILITIES

- AI Blog Generation: creates structured blog posts from high-level themes.
- AI Summarization: converts long article content into concise summaries.
- AI Flow connects specialized endpoints such as `/api/ai/generate-blog` and `/api/ai/summarize`.
- Gemini AI is integrated for intelligent content assistance.

SECURITY

- JWT for stateless authentication.
- Role-based authorization: Admin, Editor, Author, Reader.
- bcrypt.js for secure password hashing.
- Request validation and input sanitization.
- Rate limiting and structured logging for resilience.

Conclusion

What the project delivers

Secure

JWT + role matrix +
validation

Scalable

Layered Node.js
architecture

Intelligent

AI generation +
summarization

Manageable

Blogs, comments,
categories & tags

Modern REST API, MongoDB-backed data management, role-based security and Generative AI to create a

THANK YOU